

Third Section.

Transformation of the Theta-Functions.

§ 32. The principal transformations of second order of the ϑ -functions.

The two principal transformations of n -th order

$$\begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix}, \quad \begin{pmatrix} n & 0 \\ 0 & 1 \end{pmatrix}$$

transform, according to § 27, (11), the characteristic (g_1, g_2) into

$$(ng_1, g_2), \quad (g_1, ng_2),$$

that is, for odd n the characteristic remains unaltered, while for even n it passes into

$$(0, g_2) \quad \text{or} \quad (g_1, 0). \quad (1)$$

Since, according to this, the transformation of even degree behaves essentially otherwise than that of odd degree, we first consider the case $n = 2$. The first and the second principal transformation of second degree are called the Landen and the Gauss transformation.

According to § 27, (12),

$$\begin{aligned} \vartheta_{g_1, g_2} \left(u, \frac{\omega}{2} \right) &= \Theta_{g_1, 0}(u, \omega), \\ \vartheta_{g_1, g_2}(2u, 2\omega) &= \Theta_{0, g_2}(u, \omega) \end{aligned} \quad (2)$$

are Θ -functions of second order in u, ω , which can be represented according to § 21.

We first obtain the following two pairs of formulae, in which A, B are independent of u :

$$\begin{aligned} A\vartheta_{11} \left(u, \frac{\omega}{2} \right) &= \vartheta_{01}(u, \omega)\vartheta_{11}(u, \omega), \\ A\vartheta_{10} \left(u, \frac{\omega}{2} \right) &= \vartheta_{00}(u, \omega)\vartheta_{10}(u, \omega), \end{aligned} \quad (3)$$

$$\begin{aligned} B\vartheta_{11}(2u, 2\omega) &= \vartheta_{10}(u, \omega)\vartheta_{11}(u, \omega), \\ B\vartheta_{01}(2u, 2\omega) &= \vartheta_{00}(u, \omega)\vartheta_{01}(u, \omega), \end{aligned} \quad (4)$$

and in each case the second formula can be derived from the first by increasing the argument by a half-period.

Putting $u = 0$ in these equations gives

$$\begin{aligned} A\vartheta'_{11} \left(0, \frac{\omega}{2} \right) &= \vartheta_{01}\vartheta'_{11}, & 2B\vartheta'_{11}(0, 2\omega) &= \vartheta_{10}\vartheta'_{11}, \\ A\vartheta_{10} \left(0, \frac{\omega}{2} \right) &= \vartheta_{00}\vartheta_{10}, & B\vartheta_{01}(0, 2\omega) &= \vartheta_{00}\vartheta_{01}, \end{aligned} \quad (5)$$

whence, by division and with use of the relation

$$\vartheta'_{11} = \pi\vartheta_{00}\vartheta_{11}\vartheta_{01},$$

one obtains

$$\vartheta_{00} \left(0, \frac{\omega}{2} \right) \vartheta_{01} \left(0, \frac{\omega}{2} \right) = \vartheta_{01}^2, \quad (6)$$

$$2\vartheta_{00}(0, 2\omega)\vartheta_{10}(0, 2\omega) = \vartheta_{10}^2, \quad (7)$$

and, replacing ω in (6) by 2ω and ω in (7) by $\omega/2$,

$$\vartheta_{01}(0, 2\omega)^2 = \vartheta_{00}\vartheta_{01}, \quad (8)$$

$$\vartheta_{10}\left(0, \frac{\omega}{2}\right)^2 = 2\vartheta_{00}\vartheta_{10}. \quad (9)$$

From (8) and (9), the second equations (5) give

$$2A = \vartheta_{10}\left(0, \frac{\omega}{2}\right), \quad B = \vartheta_{01}(0, 2\omega),$$

and hence for the Gauss transformation one obtains

$$\begin{aligned} \vartheta_{10}\left(0, \frac{\omega}{2}\right)\vartheta_{11}\left(u, \frac{\omega}{2}\right) &= 2\vartheta_{01}(u, \omega)\vartheta_{11}(u, \omega), \\ \vartheta_{10}\left(0, \frac{\omega}{2}\right)\vartheta_{10}\left(u, \frac{\omega}{2}\right) &= 2\vartheta_{00}(u, \omega)\vartheta_{10}(u, \omega), \end{aligned} \quad (10)$$

and for the Landen transformation

$$\begin{aligned} \vartheta_{01}(0, 2\omega)\vartheta_{11}(2u, 2\omega) &= \vartheta_{10}(u, \omega)\vartheta_{11}(u, \omega), \\ \vartheta_{01}(0, 2\omega)\vartheta_{01}(2u, 2\omega) &= \vartheta_{00}(u, \omega)\vartheta_{01}(u, \omega). \end{aligned} \quad (11)$$

For each of the two transformations two ϑ -functions still remain to be expressed. These expressions may be derived from (10), (11) according to § 21, (13), but one also obtains them directly in the following way. The functions

$$\vartheta_{01}\left(u, \frac{\omega}{2}\right), \quad \vartheta_{10}(2u, 2\omega) \quad (12)$$

vanish for

$$u = \frac{\omega}{4}, \quad u = \frac{1}{4}.$$

On the other hand, from the formulae (8) of § 21, putting there $v = -\omega/4$ and $-1/4$, one obtains

$$\vartheta_{11}\left(\frac{\omega}{4}\right) = -i\vartheta_{01}\left(\frac{\omega}{4}\right), \quad \vartheta_{11}\left(\frac{1}{4}\right) = \vartheta_{10}\left(\frac{1}{4}\right),$$

and consequently the two functions (12), which are expressible linearly by two ϑ -squares, agree, apart from constant factors, with

$$\vartheta_{01}^2(u) + \vartheta_{11}^2(u), \quad \vartheta_{10}^2(u) - \vartheta_{11}^2(u).$$

The constant factors are obtained immediately by setting $u = 0$ and using the relations (6), (7):

$$\vartheta_{00}\left(0, \frac{\omega}{2}\right)\vartheta_{01}\left(u, \frac{\omega}{2}\right) = \vartheta_{01}^2(u) + \vartheta_{11}^2(u), \quad (13)$$

$$2\vartheta_{00}(0, 2\omega)\vartheta_{10}(2u, 2\omega) = \vartheta_{10}^2(u) - \vartheta_{11}^2(u). \quad (14)$$

The two final formulae are obtained from this by transforming u into $u + \frac{1}{2}$ and $u + \frac{\omega}{2}$, or also by the same method as for (13), (14):

$$\vartheta_{00}\left(0, \frac{\omega}{2}\right)\vartheta_{00}\left(u, \frac{\omega}{2}\right) = \vartheta_{00}^2(u) + \vartheta_{10}^2(u), \quad (15)$$

$$2\vartheta_{00}(0, 2\omega)\vartheta_{00}(2u, 2\omega) = \vartheta_{00}^2(u) + \vartheta_{01}^2(u). \quad (16)$$

From these one may derive manifold relations between the zero-values of the ϑ -functions. We record only the following three, of which the first two flow from (8), (9), while the last follows from the first equation (11), after replacing ω by $\omega/2$ and taking $u = 1/4$, with the fact that $\vartheta_{11}(1/4) = \vartheta_{10}(1/4)$:

$$\begin{aligned}\sqrt{\vartheta_{00}\vartheta_{01}} &= \vartheta_{01}(0, 2\omega), \\ \sqrt{\vartheta_{00}\vartheta_{10}} &= \frac{1}{\sqrt{2}} \vartheta_{10}\left(0, \frac{\omega}{2}\right), \\ \sqrt{\vartheta_{01}\vartheta_{10}} &= \vartheta_{10}\left(\frac{1}{4}, \frac{\omega}{2}\right).\end{aligned}\tag{17}$$

These formulae are of interest because they represent the square roots as single-valued functions of ω .

We make one further application of the transformation of second order to the proof of a formula needed for the transformation of odd order. Replacing ω by 2ω in the second equation (10), we have

$$2\vartheta_{00}(u, 2\omega)\vartheta_{10}(u, 2\omega) = \vartheta_{10}\vartheta_{10}(u).$$

Multiplying by the second equation (11) gives

$$2\vartheta_{01}(0, 2\omega)\vartheta_{01}(2u, 2\omega)\vartheta_{00}(u, 2\omega)\vartheta_{10}(u, 2\omega) = \vartheta_{10}\vartheta_{10}(u)\vartheta_{00}(u)\vartheta_{01}(u).$$

Dividing by the product of the two equations (7), (8),

$$2\vartheta_{00}(0, 2\omega)\vartheta_{10}(0, 2\omega)\vartheta_{01}(0, 2\omega)^2 = \vartheta_{10}^2\vartheta_{00}\vartheta_{01},$$

one obtains

$$\frac{\vartheta_{00}(u, 2\omega)\vartheta_{10}(u, 2\omega)\vartheta_{01}(2u, 2\omega)}{\vartheta_{00}(0, 2\omega)\vartheta_{10}(0, 2\omega)\vartheta_{01}(0, 2\omega)} = \frac{\vartheta_{00}(u)\vartheta_{10}(u)\vartheta_{01}(u)}{\vartheta_{00}\vartheta_{10}\vartheta_{01}}.\tag{18}$$

If now n denotes any odd integer, the function

$$\vartheta_{01}\left(\frac{v}{n}\right)$$

remains unaltered when v is increased by a multiple of n , and hence the product

$$\prod \vartheta_{01}\left(\frac{v}{n}\right),$$

when taken over a complete system of residues modulo n , is independent of the special choice of this residue system. Therefore, since $2v$ runs through such a residue system together with v ,

$$\prod_{v=1}^{n-1} \vartheta_{01}\left(\frac{v}{n}\right) = \prod_{v=1}^{n-1} \vartheta_{01}\left(\frac{2v}{n}\right).\tag{19}$$

Putting $u = v/n$ in (18), forming the product, and applying (19) to the last factor on the left, one obtains that the product

$$\frac{\prod_{v=1}^{n-1} \vartheta_{00}\left(\frac{v}{n}\right)\vartheta_{10}\left(\frac{v}{n}\right)\vartheta_{01}\left(\frac{v}{n}\right)}{\vartheta_{00}^{n-1}\vartheta_{10}^{n-1}\vartheta_{01}^{n-1}}\tag{20}$$

remains unaltered when ω is replaced by 2ω .

The values of v occurring in (20) can be arranged in pairs

$$v, n-v, \quad v = 1, 2, \dots, \frac{n-1}{2},$$

and, since

$$\vartheta_{00}\left(\frac{v}{n}\right) = \vartheta_{00}\left(\frac{n-v}{n}\right), \quad \vartheta_{10}\left(\frac{v}{n}\right) = -\vartheta_{10}\left(\frac{n-v}{n}\right), \quad \vartheta_{01}\left(\frac{v}{n}\right) = \vartheta_{01}\left(\frac{n-v}{n}\right),$$

(20), except for the sign, agrees with the square of

$$\frac{\prod_{v=1}^{(n-1)/2} \vartheta_{00}\left(\frac{v}{n}\right) \vartheta_{10}\left(\frac{v}{n}\right) \vartheta_{01}\left(\frac{v}{n}\right)}{\vartheta_{00}^{(n-1)/2} \vartheta_{10}^{(n-1)/2} \vartheta_{01}^{(n-1)/2}}. \quad (21)$$

The latter quotient, which is a continuous non-vanishing function of ω as long as the imaginary part of ω is positive, therefore also remains unchanged when ω is replaced by 2ω , and hence by $4\omega, 8\omega, \dots$. Its value can be determined by allowing the imaginary part of ω to become infinite, that is by taking $q = 0$. But for $q = 0$ one has, by §25,

$$\vartheta_{00}(v) = 1, \quad \vartheta_{01}(v) = 1, \quad \frac{\vartheta_{10}(v)}{\vartheta_{10}} = \cos \pi v,$$

and hence the value of (21) is

$$\prod_{v=1}^{(n-1)/2} \cos \frac{v\pi}{n}. \quad (22)$$

Since here $v\pi/n$ is less than $\frac{1}{2}\pi$, this product has a positive value. But

$$\cos \frac{v\pi}{n} = -\cos \frac{(n-v)\pi}{n},$$

and by Vol. I, §144,

$$2^{(n-1)/2} \prod_{v=1}^{(n-1)/2} \cos \frac{v\pi}{n} = 1.$$

Thus the formula is proved:

$$2^{(n-1)/2} \prod_{v=1}^{(n-1)/2} \vartheta_{00}\left(\frac{v}{n}\right) \vartheta_{10}\left(\frac{v}{n}\right) \vartheta_{01}\left(\frac{v}{n}\right) = \vartheta_{00}^{(n-1)/2} \vartheta_{10}^{(n-1)/2} \vartheta_{01}^{(n-1)/2}. \quad (23)$$

With regard to the formula Vol. I, §145, (3), one can give this relation also the form

$$2^{(n-1)/2} \prod_{v=1}^{(n-1)/2} \vartheta_{00}\left(\frac{2v}{n}\right) \vartheta_{10}\left(\frac{2v}{n}\right) \vartheta_{01}\left(\frac{2v}{n}\right) = (-1)^{(n^2-1)/8} \vartheta_{00}^{(n-1)/2} \vartheta_{10}^{(n-1)/2} \vartheta_{01}^{(n-1)/2}. \quad (24)$$

§ 33. The principal transformations of odd order.

The formula proved last is useful to us in carrying out the transformation of odd order n . We first consider the first principal transformation. According to §27, (12),

$$\vartheta_{11}(nu, n\omega) \quad (1)$$

is a Θ_{11} -function of n -th order in u and ω , and it can readily be formed from its zeros, of which there are n in the period-parallelogram. The zeros of (1) are the values

$$u = \frac{v + \mu n\omega}{n} = \frac{v}{n} + \mu\omega,$$

where v and μ are integers, and all incongruent values among these are obtained when μ is fixed and v runs through a complete system of residues modulo n . We choose the system of residues

$$0, \pm 1, \pm 2, \dots, \pm \frac{n-1}{2},$$

and hence, if C denotes a factor independent of u , obtain

$$C\vartheta_{11}(nu, n\omega) = \vartheta_{11}(u) \prod_{v=1}^{(n-1)/2} \vartheta_{11}\left(\frac{v}{n} + u\right) \vartheta_{11}\left(\frac{v}{n} - u\right), \quad (2)$$

a formula which, by § 22, (4), can also be expressed in the following way through the functions $\vartheta_{11}(u), \vartheta_{01}(u)$:

$$C\vartheta_{01}^{n-1}\vartheta_{11}(nu, n\omega) = \vartheta_{11}(u) \prod_{v=1}^{(n-1)/2} \left[\vartheta_{11}^2\left(\frac{v}{n}\right) \vartheta_{01}^2(u) - \vartheta_{01}^2\left(\frac{v}{n}\right) \vartheta_{11}^2(u) \right].$$

We replace u by

$$u + \frac{1}{2}, \quad u + \frac{\omega}{2}, \quad u + \frac{1+\omega}{2},$$

and obtain from (8) of § 21:

$$\begin{aligned} C\vartheta_{10}(nu, n\omega) &= \vartheta_{10}(u) \prod_{v=1}^{(n-1)/2} \vartheta_{10}\left(\frac{v}{n} + u\right) \vartheta_{10}\left(\frac{v}{n} - u\right), \\ C\vartheta_{01}(nu, n\omega) &= \vartheta_{01}(u) \prod_{v=1}^{(n-1)/2} \vartheta_{01}\left(\frac{v}{n} + u\right) \vartheta_{01}\left(\frac{v}{n} - u\right), \\ C\vartheta_{00}(nu, n\omega) &= \vartheta_{00}(u) \prod_{v=1}^{(n-1)/2} \vartheta_{00}\left(\frac{v}{n} + u\right) \vartheta_{00}\left(\frac{v}{n} - u\right). \end{aligned} \quad (3)$$

Putting $u = 0$ and using $\vartheta'_{11} = \pi\vartheta_{00}\vartheta_{01}\vartheta_{10}$ gives

$$C \prod_{v=1}^{(n-1)/2} \vartheta_{11}\left(\frac{v}{n}\right) = \sqrt{n} \prod_{v=1}^{(n-1)/2} \vartheta_{00}\left(\frac{v}{n}\right) \vartheta_{01}\left(\frac{v}{n}\right) \vartheta_{10}\left(\frac{v}{n}\right), \quad (4)$$

or, with use of (23) of the preceding paragraph,

$$C 2^{(n-1)/2} \prod_{v=1}^{(n-1)/2} \vartheta_{11}\left(\frac{v}{n}\right) = \sqrt{n} \vartheta_{00}^{(n-1)/2} \vartheta_{10}^{(n-1)/2} \vartheta_{01}^{(n-1)/2}. \quad (5)$$

The sign is obtained from the fact, following from one of the formulae (3), that $C = 1$ for $q = 0$.

After this determination of C , the formulae (2), (3) may be written as

$$\begin{aligned} \sqrt{n} \vartheta_{11}(nu, n\omega) \vartheta_{00}^{(n-1)/2} \vartheta_{10}^{(n-1)/2} \vartheta_{01}^{(n-1)/2} \\ = 2^{(n-1)/2} \vartheta_{11}(u) \prod_{v=1}^{(n-1)/2} \vartheta_{11}\left(\frac{v}{n}\right) \vartheta_{11}\left(\frac{v}{n} + u\right) \vartheta_{11}\left(\frac{v}{n} - u\right), \end{aligned} \quad (6)$$

$$\begin{aligned} \sqrt{n} \vartheta_{10}(nu, n\omega) \vartheta_{00}^{(n-1)/2} \vartheta_{10}^{(n-1)/2} \vartheta_{01}^{(n-1)/2} \\ = 2^{(n-1)/2} \vartheta_{10}(u) \prod_{v=1}^{(n-1)/2} \vartheta_{11}\left(\frac{v}{n}\right) \vartheta_{01}\left(\frac{v}{n} + u\right) \vartheta_{01}\left(\frac{v}{n} - u\right), \end{aligned} \quad (7)$$

$$\begin{aligned} \sqrt{n} \vartheta_{01}(nu, n\omega) \vartheta_{00}^{(n-1)/2} \vartheta_{10}^{(n-1)/2} \vartheta_{01}^{(n-1)/2} \\ = 2^{(n-1)/2} \vartheta_{01}(u) \prod_{v=1}^{(n-1)/2} \vartheta_{11}\left(\frac{v}{n}\right) \vartheta_{01}\left(\frac{v}{n} + u\right) \vartheta_{01}\left(\frac{v}{n} - u\right), \end{aligned} \quad (8)$$

$$\begin{aligned} \sqrt{n} \vartheta_{00}(nu, n\omega) \vartheta_{00}^{(n-1)/2} \vartheta_{10}^{(n-1)/2} \vartheta_{01}^{(n-1)/2} \\ = 2^{(n-1)/2} \vartheta_{00}(u) \prod_{v=1}^{(n-1)/2} \vartheta_{11}\left(\frac{v}{n}\right) \vartheta_{00}\left(\frac{v}{n} + u\right) \vartheta_{00}\left(\frac{v}{n} - u\right). \end{aligned} \quad (9)$$

§ 34. The functions $\eta(\omega)$, $f(\omega)$, $f_1(\omega)$, $f_2(\omega)$.

The functions $\eta(\omega)$, $f(\omega)$, $f_1(\omega)$, $f_2(\omega)$ were already mentioned in § 24. There they arose almost of themselves as single-valued functions of ω in the representation of the ϑ -functions by infinite products, whereas they did not arise in the second representation by infinite series. Connected with this is the remarkable fact that many of our formulae, for example the formulae (17) of § 32 belonging to the transformation of second order, or the formula (23) of § 32, may easily be verified by the infinite products, but only with difficulty, or not at all, by the infinite series. It was therefore of interest to us to derive these results, without using either of those expressions, from the theory of transformation; and from this same source we shall now obtain the functions $\eta(\omega)$, $f(\omega)$, $f_1(\omega)$, $f_2(\omega)$ and their fundamental properties.

Formula (6) of the preceding paragraph gives, for $n = 3$,

$$\sqrt{3} \vartheta_{11}(3u, 3\omega) \vartheta_{00} \vartheta_{10} \vartheta_{01} = 2 \vartheta_{11} \left(\frac{1}{3} \right) \vartheta_{11}(u) \vartheta_{11} \left(\frac{1}{3} + u \right) \vartheta_{11} \left(\frac{1}{3} - u \right),$$

and if one differentiates and then sets $u = 0$, using (4),

$$3\sqrt{3} \vartheta'_{11}(0, 3\omega) = 2\pi \vartheta_{11} \left(\frac{1}{3} \right)^3,$$

or, replacing ω by $\omega/3$,

$$3\sqrt{3} \vartheta'_{11} = 2\pi \left[\vartheta_{11} \left(\frac{1}{3}, \frac{\omega}{3} \right) \right]^3.$$

Thus, putting

$$\eta(\omega) = \frac{1}{\sqrt{3}} \vartheta_{11} \left(\frac{1}{3}, \frac{\omega}{3} \right), \quad (1)$$

one obtains, in agreement with the notation of § 24, (9),

$$\vartheta'_{11} = \pi \vartheta_{00} \vartheta_{01} \vartheta_{10} = 2\pi \eta(\omega)^3. \quad (2)$$

From § 25, (4), one obtains for $\eta(\omega)$ the series expansion

$$\eta(\omega) = q^{1/12} \sum_{\nu=-\infty}^{\infty} (-1)^\nu q^{3\nu^2+\nu}. \quad (3)$$

The function $\eta(\omega)$ is real for purely imaginary ω (real q), and for infinitely large ω (that is, for vanishing q),

$$q^{-1/12} \eta(\omega) = 1.$$

Hence, applying to (2) the formulae § 31, (6), (11), one finds for the linear fundamental transformations of $\eta(\omega)$

$$\eta(\omega + 1) = e^{\pi i/12} \eta(\omega), \quad (4)$$

$$\eta \left(-\frac{1}{\omega} \right) = \sqrt{-i\omega} \eta(\omega), \quad (5)$$

of which the first also follows immediately from the series representation (3), while the second is not so easily proved by a direct transformation of the series.

We now pass to the consideration of the two principal transformations of second order of the η -function. From § 32, (11), by differentiating and then setting $u = 0$, one obtains

$$2\vartheta_{01}(0, 2\omega) \eta(2\omega)^3 = \vartheta_{10} \eta(\omega)^3,$$

and therefore, after squaring and using § 32, (8),

$$4\vartheta_{00} \vartheta_{10} \vartheta_{01} \eta(2\omega)^6 = \vartheta_{10}^3 \eta(\omega)^6.$$

With use of (2) and extraction of the cube root this gives

$$2\eta(2\omega)^2 = \vartheta_{10}\eta(\omega). \quad (6)$$

Replacing ω in (6) by $\omega/2$ gives

$$2\eta(\omega)^2 = \vartheta_{10} \left(0, \frac{\omega}{2}\right) \eta\left(\frac{\omega}{2}\right),$$

and, after squaring and applying § 32, (9),

$$2\eta(\omega)^4 = \eta\left(\frac{\omega}{2}\right)^2 \vartheta_{00}\vartheta_{10}.$$

Multiplication on both sides by ϑ_{01} gives, according to (2),

$$\eta\left(\frac{\omega}{2}\right)^2 = \vartheta_{01}\eta(\omega). \quad (7)$$

By changing ω into $\omega + 1$, using (4) and § 31, (6), one obtains

$$e^{-\pi i/12} \eta\left(\frac{\omega+1}{2}\right)^2 = \vartheta_{00}\eta(\omega). \quad (8)$$

Accordingly we introduce the three functions

$$f(\omega) = \frac{e^{-\pi i/24} \eta\left(\frac{\omega+1}{2}\right)}{\eta(\omega)}, \quad f_1(\omega) = \frac{\eta\left(\frac{\omega}{2}\right)}{\eta(\omega)}, \quad f_2(\omega) = \frac{\sqrt{2}\eta(2\omega)}{\eta(\omega)}. \quad (9)$$

Then, by (6), (7), (8),

$$\begin{aligned} \vartheta_{00} &= \eta(\omega)f(\omega)^2, \\ \vartheta_{01} &= \eta(\omega)f_1(\omega)^2, \\ \vartheta_{10} &= \eta(\omega)f_2(\omega)^2. \end{aligned} \quad (10)$$

It follows from (8) and (9) that $f(\omega), f_1(\omega), f_2(\omega)$ are real and positive for purely imaginary ω . Equation (10) agrees with § 24, (10), and the functions f, f_1, f_2 introduced there are the same as these. From (10) the relations [(2) and § 21, (14)] follow easily:

$$f(\omega)^8 = f_1(\omega)^8 + f_2(\omega)^8, \quad \sqrt{2} = f(\omega)f_1(\omega)f_2(\omega). \quad (11)$$

With the aid of these formulae one can, by square roots, express any one of the three functions $f(\omega)^8, f_1(\omega)^8, f_2(\omega)^8$ in terms of any other. This is furnished by the following formulae derived from (11):

$$\begin{aligned} [f_1(\omega)^8 - f_2(\omega)^8]^2 &= f(\omega)^{16} - \frac{64}{f(\omega)^8}, \\ [f(\omega)^8 + f_2(\omega)^8]^2 &= f_1(\omega)^{16} + \frac{64}{f_1(\omega)^8}, \\ [f(\omega)^8 + f_1(\omega)^8]^2 &= f_2(\omega)^{16} + \frac{64}{f_2(\omega)^8}. \end{aligned} \quad (12)$$

For the linear fundamental transformations of the functions f , one obtains first from (4) and (9):

$$\begin{aligned} f(\omega + 1) &= e^{-\pi i/24} f_1(\omega), \\ f_1(\omega + 1) &= e^{-\pi i/24} f(\omega), \\ f_2(\omega + 1) &= e^{\pi i/12} f_2(\omega). \end{aligned} \quad (13)$$

Furthermore, from (5) and the two last equations (9),

$$f_1\left(-\frac{1}{\omega}\right) = f_2(\omega), \quad f_2\left(-\frac{1}{\omega}\right) = f_1(\omega), \quad (14)$$

and using these in the second equation (11),

$$f\left(-\frac{1}{\omega}\right) = f(\omega). \quad (15)$$

For the transformation of second order, one obtains immediately from the two last equations (9)

$$f_1(2\omega)f_2(\omega) = \sqrt{2}, \quad (16)$$

from which, by means of (12), it follows also that

$$f_2(\omega)^4[f(2\omega)^8 + f_2(2\omega)^8] = 2[f(\omega)^8 + f_1(\omega)^8]. \quad (17)$$

Putting in (16), according to (13), (14),

$$f_1(2\omega) = e^{\pi i/24} f(2\omega - 1), \quad f_2(\omega) = e^{\pi i/24} f\left(1 - \frac{1}{\omega}\right),$$

one obtains

$$f\left(1 - \frac{1}{\omega}\right) f(2\omega - 1) = \sqrt{2},$$

and, putting $2\omega - 1$ equal to a new ω ,

$$f(\omega) f\left(\frac{\omega - 1}{\omega + 1}\right) = \sqrt{2}. \quad (18)$$

Replacing ω in (16) by $\omega/2$ and $(\omega + 1)/2$ gives

$$\begin{aligned} f_1(\omega) f_2\left(\frac{\omega}{2}\right) &= \sqrt{2}, \\ f(\omega) f_2\left(\frac{\omega + 1}{2}\right) &= e^{\pi i/24} \sqrt{2}. \end{aligned} \quad (19)$$

Finally, we set down the formulae for the principal transformation of odd order n of the functions η, f . For $\eta(n\omega)$ one obtains easily, if in formulae (6) of § 33 one differentiates and sets $u = 0$, and then extracts the cube root,

$$\sqrt{n} \eta(n\omega) \eta(\omega)^{(n-3)/2} = \prod_{v=1}^{(n-1)/2} \vartheta_{11}\left(\frac{v}{n}\right). \quad (20)$$

Putting $u = 0$ in (7), (8), (9) of § 33, using the relations (10) and (20), and extracting the square root, one finds

$$\begin{aligned} f(n\omega) \eta(\omega)^{(n-1)/2} &= f(\omega) \prod_{v=1}^{(n-1)/2} \vartheta_{00}\left(\frac{v}{n}\right), \\ f_1(n\omega) \eta(\omega)^{(n-1)/2} &= f_1(\omega) \prod_{v=1}^{(n-1)/2} \vartheta_{01}\left(\frac{v}{n}\right), \\ f_2(n\omega) \eta(\omega)^{(n-1)/2} &= f_2(\omega) \prod_{v=1}^{(n-1)/2} \vartheta_{10}\left(\frac{v}{n}\right). \end{aligned} \quad (21)$$

The signs here are obtained from the assumption of infinitely small q .

§ 35. The Weierstrass σ -function.

By the general linear substitution

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \quad \alpha\delta - \beta\gamma = 1,$$

applied to the variables ω_1, ω_2 , every t -function passes again into a t -function, while the characteristic is changed. If (g_1, g_2) is the characteristic of the original function and (g'_1, g'_2) that of the transformed function, then, by § 27, (11),

$$(g_1, g_2) = (\delta g'_1 - \beta g'_2 - \beta\delta, -\gamma g'_1 + \alpha g'_2 - \alpha\gamma).$$

Solving the two linear equations contained in this for g'_1, g'_2 , and observing that $\alpha\beta(\gamma + \delta + 1)$ and $\gamma\delta(\alpha + \beta + 1)$ are necessarily even numbers, and that a characteristic is not changed if its elements are changed by multiples of 2, one may put equivalently

$$(g'_1, g'_2) = (\alpha g_1 + \beta g_2 + \alpha\beta, \gamma g_1 + \delta g_2 + \gamma\delta). \quad (1)$$

Thus, apart from exponential factors, the ϑ -functions pass into one another. It is, however, important to form a function which is absolutely invariant under linear transformations, and such a function is the σ -function introduced into the theory by Weierstrass; we now pass to its definition.

If formula (1) is applied to the four principal characteristics $(0, 0), (0, 1), (1, 0), (1, 1)$, these pass, in order, into $(\alpha\beta, \gamma\delta), ((\alpha+1)\beta, (\gamma+1)\delta), (\alpha(\beta+1), \gamma(\delta+1)), ((\alpha+1)(\beta+1)-1, (\gamma+1)(\delta+1)-1) = (1, 1)$. Thus only the last characteristic, $(1, 1)$, remains unchanged under all linear transformations, since neither α and β nor γ and δ can both be even. Let, therefore, $t(u, \omega_1, \omega_2)$ be a t -function with characteristic $(1, 1)$, which by § 17, (12), must vanish for $u = 0$.

If

$$(\omega'_1, \omega'_2) = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (\omega_1, \omega_2),$$

then, by our transformation principle,

$$t(u, \omega_1, \omega_2), \quad t(u, \omega'_1, \omega'_2)$$

are related T -functions of first order, and hence, if C, λ, μ are quantities independent of u ,

$$t(u, \omega'_1, \omega'_2) = C e^{\lambda u^2 + \mu u} t(u, \omega_1, \omega_2). \quad (2)$$

It follows by logarithmic differentiation that

$$2\lambda u + \mu = \frac{d \log t(u, \omega'_1, \omega'_2)}{du} - \frac{d \log t(u, \omega_1, \omega_2)}{du}. \quad (3)$$

Expanding in powers of u by Taylor's theorem, and denoting by t', t'', t''' the first, second, and third derivatives of t with respect to u for $u = 0$, one has

$$\frac{d \log t(u, \omega_1, \omega_2)}{du} = \frac{1}{u} + \frac{t''}{2t'} + u \left(\frac{t'''}{3t'} - \frac{t''^2}{4t'^2} \right) + \cdots,$$

from which, comparing with (3), it follows that λ and μ can be written in the form

$$\lambda = \varphi(\omega'_1, \omega'_2) - \varphi(\omega_1, \omega_2), \quad \mu = \psi(\omega'_1, \omega'_2) - \psi(\omega_1, \omega_2),$$

where

$$\varphi(\omega_1, \omega_2) = \frac{t'''}{6t'} - \frac{t''^2}{8t'^2}, \quad \psi(\omega_1, \omega_2) = \frac{t''}{2t'}. \quad (4)$$

If one further determines the factor C in (2) by the special value $u = 0$, then

$$C = \frac{t'(\omega'_1, \omega'_2)}{t'(\omega_1, \omega_2)}. \quad (5)$$

Consequently formula (2) can be expressed in the following theorem: the function

$$\sigma(u, \omega_1, \omega_2) = e^{-u^2 \left(\frac{t'''}{6t'} - \frac{t''^2}{8t'^2} \right) - u \frac{t''}{2t'}} \frac{t(u, \omega_1, \omega_2)}{t'} \quad (6)$$

remains unchanged when ω_1, ω_2 are replaced by ω'_1, ω'_2 , that is, when any linear transformation is applied:

$$\sigma(u, \omega'_1, \omega'_2) = \sigma(u, \omega_1, \omega_2). \quad (7)$$

The function defined by (6) remains, as a simple calculation shows, unchanged if t is replaced by any related t -function.

The function $t(u)$ has, by assumption, the characteristic $(1, 1)$, and its zeros are therefore congruent to 0. Hence $t(-u)$ has the same zeros and also the same character as $t(u)$. The two functions differ only by an exponential factor, and one easily obtains, determining the constant factor from $u = 0$,

$$t(-u) = -e^{2\pi i \mu u} t(u),$$

where μ is a constant. Differentiating this equation twice with respect to u and then putting $u = 0$ gives

$$2\pi i \mu = -\frac{t''}{t'},$$

and hence by (6) one obtains that the function $\sigma(u)$ satisfies

$$\sigma(-u) = -\sigma(u), \quad (8)$$

that is, is an odd function of u .

Since $\sigma(u)$ is a t -function of first order, it suffices to satisfy the two conditions

$$\sigma(u + \omega_1) = c_1 e^{\eta_1(2u + \omega_1)} \sigma(u), \quad \sigma(u + \omega_2) = c_2 e^{\eta_2(2u + \omega_2)} \sigma(u),$$

where c_1, c_2, η_1, η_2 are constants. The function $\sigma(u)$ vanishes for $u = 0$, but not for $u = \omega_1/2$ or $u = \omega_2/2$; if, therefore, in the preceding formulae one puts $u = -\omega_1/2$ and $u = -\omega_2/2$, respectively, then by (8) one obtains $c_1 = -1, c_2 = -1$, and hence the formulae

$$\begin{aligned} \sigma(u + \omega_1) &= -e^{\eta_1(2u + \omega_1)} \sigma(u), \\ \sigma(u + \omega_2) &= -e^{\eta_2(2u + \omega_2)} \sigma(u). \end{aligned} \quad (9)$$

The quantities η_1, η_2 here introduced are functions of ω_1, ω_2 , satisfying the equation

$$\eta_1 \omega_2 - \eta_2 \omega_1 = \pi i, \quad (10)$$

according to the relation § 17, (10), in which $\pi i a_1, \pi i a_2, m$ are to be replaced by $-\eta_1, -\eta_2, 1$.

By repeated application of (9), if a, b are integers, one obtains [cf. § 17, (16)]

$$\sigma(u + a\omega_1 + b\omega_2) = (-1)^{a+b+ab} e^{(a\eta_1 + b\eta_2)(2u + a\omega_1 + b\omega_2)} \sigma(u). \quad (11)$$

When the function $\sigma(u)$, as given by formula (6), is expanded in powers of u , one finds that not only the second, but also the third power of u does not occur in the expansion; hence

$$\sigma(0) = 0, \quad \sigma'(0) = 1, \quad \sigma''(0) = 0, \quad \sigma'''(0) = 0. \quad (12)$$

If the linear substitution

$$(\omega'_1, \omega'_2) = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (\omega_1, \omega_2) \quad (13)$$

is applied to ω_1, ω_2 , then the quantities η_1, η_2 undergo the corresponding substitution

$$(\eta'_1, \eta'_2) = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (\eta_1, \eta_2), \quad (14)$$

as follows immediately from the relations (7) and (11).

Logarithmically differentiating the formulae (9) with respect to u gives

$$\begin{aligned} \frac{\sigma'(u + \omega_1)}{\sigma(u + \omega_1)} &= \frac{\sigma'(u)}{\sigma(u)} + 2\eta_1, \\ \frac{\sigma'(u + \omega_2)}{\sigma(u + \omega_2)} &= \frac{\sigma'(u)}{\sigma(u)} + 2\eta_2, \end{aligned} \quad (15)$$

and by putting in the first equation $u = -\omega_1/2$ and in the second $u = -\omega_2/2$, and observing that $\sigma(u)$ is odd and hence $\sigma'(u)$ is even,

$$\eta_1 = \frac{\sigma'(\omega_1/2)}{\sigma(\omega_1/2)}, \quad \eta_2 = \frac{\sigma'(\omega_2/2)}{\sigma(\omega_2/2)}. \quad (16)$$

§ 36. The functions $\sigma_{00}, \sigma_{01}, \sigma_{10}$.

It remains to obtain the analogous results for the remaining characteristics. We proceed from the following observation. If $x_1, x_2; y_1, y_2$ are any two pairs of quantities which are transformed by the same linear substitution S into $x'_1, x'_2; y'_1, y'_2$, so that

$$(x'_1, x'_2) = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (x_1, x_2), \quad (y'_1, y'_2) = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (y_1, y_2),$$

then also

$$x_1 y_2 - x_2 y_1 = x'_1 y'_2 - x'_2 y'_1,$$

as follows from the multiplication of determinants. Hence, whatever x_1, x_2 may be, by (7) of § 35,

$$\sigma(u + x_2 \omega_1 - x_1 \omega_2, \omega_1, \omega_2) = \sigma(u + x'_2 \omega'_1 - x'_1 \omega'_2, \omega'_1, \omega'_2). \quad (1)$$

This function, by formula (11) of the preceding paragraph, changes when x_1 and x_2 are changed by integers. This change can be avoided if one considers instead the function

$$e^{-2(\eta_1 x_2 - \eta_2 x_1)u} \frac{\sigma(u + x_2 \omega_1 - x_1 \omega_2)}{\sigma(x_2 \omega_1 - x_1 \omega_2)},$$

which we shall for the moment denote by $\sigma(u, x_1, x_2, \omega_1, \omega_2)$. This function satisfies the conditions

$$\sigma(u, x'_1, x'_2, \omega'_1, \omega'_2) = \sigma(u, x_1, x_2, \omega_1, \omega_2), \quad (2)$$

$$\begin{aligned} \sigma(u, x_1 + 1, x_2, \omega_1, \omega_2) &= \sigma(u, x_1, x_2, \omega_1, \omega_2), \\ \sigma(u, x_1, x_2 + 1, \omega_1, \omega_2) &= \sigma(u, x_1, x_2, \omega_1, \omega_2), \end{aligned} \quad (3)$$

$$\begin{aligned} \sigma(u + \omega_1) &= -e^{-2\pi i x_2} e^{\eta_1(2u + \omega_1)} \sigma(u), \\ \sigma(u + \omega_2) &= -e^{-2\pi i x_1} e^{\eta_2(2u + \omega_2)} \sigma(u), \end{aligned} \quad (4)$$

and thus remains unchanged when x_1 and x_2 are changed by integers.

Restricting ourselves again to the principal characteristics, we put

$$(x_1, x_2) = \left(-\frac{1}{2}, \frac{1}{2}\right), \quad \left(0, \frac{1}{2}\right), \quad \left(-\frac{1}{2}, 0\right),$$

and so obtain the following three functions:

$$\begin{aligned} \sigma_{00}(u) &= \sigma\left(u, -\frac{1}{2}, \frac{1}{2}, \omega_1, \omega_2\right) = e^{-(\eta_1 + \eta_2)u} \frac{\sigma\left(u + \frac{\omega_1 + \omega_2}{2}\right)}{\sigma\left(\frac{\omega_1 + \omega_2}{2}\right)}, \\ \sigma_{10}(u) &= \sigma\left(u, 0, \frac{1}{2}, \omega_1, \omega_2\right) = e^{-\eta_1 u} \frac{\sigma\left(u + \frac{\omega_1}{2}\right)}{\sigma\left(\frac{\omega_1}{2}\right)}, \\ \sigma_{01}(u) &= \sigma\left(u, -\frac{1}{2}, 0, \omega_1, \omega_2\right) = e^{-\eta_2 u} \frac{\sigma\left(u + \frac{\omega_2}{2}\right)}{\sigma\left(\frac{\omega_2}{2}\right)}. \end{aligned} \quad (5)$$

The function $\sigma(u)$ itself can correspondingly be denoted by $\sigma_{11}(u)$.

For these functions, (4) gives the characteristic period equations

$$\begin{aligned} \sigma_{g_1, g_2}(u + \omega_1) &= (-1)^{g_1} e^{\eta_1(2u + \omega_1)} \sigma_{g_1, g_2}(u), \\ \sigma_{g_1, g_2}(u + \omega_2) &= (-1)^{g_2} e^{\eta_2(2u + \omega_2)} \sigma_{g_1, g_2}(u), \\ \sigma_{g_1, g_2}(u + a\omega_1 + b\omega_2) &= (-1)^{g_1 a + g_2 b + ab} e^{(a\eta_1 + b\eta_2)(2u + a\omega_1 + b\omega_2)} \sigma_{g_1, g_2}(u). \end{aligned} \quad (6)$$

By application of a linear transformation the three functions $\sigma_{00}, \sigma_{01}, \sigma_{10}$ are permuted among one another, as formula (2) shows, or § 35, (1). According to this permutation the linear transformations decompose into six classes, the first of which comprises all transformations leaving the functions $\sigma_{00}, \sigma_{01}, \sigma_{10}$ unchanged. These are characterized by α, δ being odd and β, γ even, which we write briefly as

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{2}.$$

Accordingly the six classes of linear transformations are characterized as follows:

$$\begin{aligned} \text{I. } & \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{2} \quad (00, 01, 10), \\ \text{II. } & \equiv \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \pmod{2} \quad (00, 10, 01), \\ \text{III. } & \equiv \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix} \pmod{2} \quad (10, 00, 01), \\ \text{IV. } & \equiv \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \pmod{2} \quad (10, 01, 00), \\ \text{V. } & \equiv \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \pmod{2} \quad (01, 00, 10), \\ \text{VI. } & \equiv \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \pmod{2} \quad (01, 10, 00), \end{aligned} \quad (7)$$

where in the last column the corresponding permutation of the characteristics 00, 01, 10 is indicated.

The transformations of the first class form a group $\mathfrak{A}$ contained in the group $\mathfrak{L}$ of all linear substitutions, that is, a divisor of the group $\mathfrak{L}$ (§ 28). If

$$\alpha_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \alpha_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \alpha_3 = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix},$$

$$\alpha_4 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \alpha_5 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad \alpha_6 = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix},$$

then $\alpha_1\mathfrak{A}, \alpha_2\mathfrak{A}, \alpha_3\mathfrak{A}, \alpha_4\mathfrak{A}, \alpha_5\mathfrak{A}, \alpha_6\mathfrak{A}$ are the cosets relative to $\mathfrak{A}$, and

$$\mathfrak{L} = \alpha_1\mathfrak{A} + \alpha_2\mathfrak{A} + \alpha_3\mathfrak{A} + \alpha_4\mathfrak{A} + \alpha_5\mathfrak{A} + \alpha_6\mathfrak{A},$$

and $\mathfrak{A}$ is a divisor of $\mathfrak{L}$ of finite index 6:

$$(\mathfrak{L}, \mathfrak{A}) = 6 \quad (\text{Vol. II, § 2}).$$

The totality $\alpha_1\mathfrak{A} + \alpha_2\mathfrak{A}$ is likewise a group $\mathfrak{A}'$, and

$$\mathfrak{L} = \alpha_1\mathfrak{A}' + \alpha_3\mathfrak{A}' + \alpha_4\mathfrak{A}', \quad (\mathfrak{L}, \mathfrak{A}') = 3.$$

Just as all linear transformations can be derived from the two fundamental transformations, so the transformations of the first class can be derived by repeated application of

$$\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix},$$

which may be proved in the manner of § 30.

§ 37. Representation of the σ -functions by ϑ -functions.

In order to represent the function $\sigma(u)$ as a ϑ -function, one can simply apply formula (6) of § 35 to the function

$$t(u) = \vartheta_{11} \left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1} \right),$$

that is, put

$$\sigma(u, \omega_1, \omega_2) = \omega_1 e^{-\frac{u^2}{6\omega_1^2} \frac{\vartheta_{11}'''}{\vartheta_{11}'}} \frac{\vartheta_{11} \left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1} \right)}{\vartheta_{11}'}. \quad (1)$$

In consequence of the differential equation § 20, (4), however,

$$\vartheta_{11}''' = 4\pi i \frac{d\vartheta_{11}'}{d\omega},$$

and hence, by the definition of the function $\eta(\omega)$ in § 34, (2),

$$\frac{\vartheta_{11}'''}{\vartheta_{11}'} = 4\pi i \frac{d \log \vartheta_{11}'}{d\omega} = 12\pi i \frac{d \log \eta(\omega)}{d\omega}. \quad (2)$$

Logarithmic differentiation of (1), using (2), gives

$$\frac{d \log \sigma(u)}{du} = -\frac{4\pi i u}{\omega_1^2} \frac{d \log \eta(\omega)}{d\omega} + \frac{d \log \vartheta_{11} \left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1} \right)}{du},$$

and, putting herein $u = \omega_1/2, \omega_2/2$, and applying the formulae (8) of § 21 together with § 35, (16),

$$\begin{aligned} \eta_1 &= -\frac{2\pi i}{\omega_1} \frac{d \log \eta(\omega)}{d\omega}, \\ \eta_2 &= -\frac{2\pi i \omega_2}{\omega_1^2} \frac{d \log \eta(\omega)}{d\omega} - \frac{\pi i}{\omega_1}. \end{aligned} \quad (3)$$

Consequently one can put for σ :

$$\sigma(u) = \omega_1 e^{\eta_1 u^2 / \omega_1} \frac{\vartheta_{11}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)}{\vartheta'_{11}}, \quad (4)$$

and for the three functions $\sigma_{00}, \sigma_{01}, \sigma_{10}$, by § 36, (5), when one determines a constant factor from

$$\sigma_{00}(0) = 1, \quad \sigma_{01}(0) = 1, \quad \sigma_{10}(0) = 1, \quad (5)$$

one obtains

$$\begin{aligned} \sigma_{00}(u) &= e^{\eta_1 u^2 / \omega_1} \frac{\vartheta_{00}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)}{\vartheta_{00}}, \\ \sigma_{01}(u) &= e^{\eta_1 u^2 / \omega_1} \frac{\vartheta_{01}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)}{\vartheta_{01}}, \\ \sigma_{10}(u) &= e^{\eta_1 u^2 / \omega_1} \frac{\vartheta_{10}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)}{\vartheta_{10}}. \end{aligned} \quad (6)$$

The functions $\sigma_{00}(u), \sigma_{01}(u), \sigma_{10}(u)$ are even functions of u , and by twice differentiating the logarithms of (6) one further obtains:

$$\sigma''_{00}(0) + \sigma''_{01}(0) + \sigma''_{10}(0) = 0. \quad (7)$$

The expressions (4), (6) show at first glance an important property of the σ -functions, namely that $\sigma_{00}, \sigma_{01}, \sigma_{10}$ depend only on the ratios $u : \omega_1 : \omega_2$, while the same is true of σ , apart from the factor ω_1 . Thus $\sigma, \sigma_{00}, \sigma_{01}, \sigma_{10}$ are homogeneous functions of the three variables u, ω_1, ω_2 , the first of order one and the other three of order zero, or, in symbols, if λ denotes an arbitrary factor,

$$\begin{aligned} \sigma(\lambda u, \lambda \omega_1, \lambda \omega_2) &= \lambda \sigma(u, \omega_1, \omega_2), \\ \sigma_{00}(\lambda u, \lambda \omega_1, \lambda \omega_2) &= \sigma_{00}(u, \omega_1, \omega_2), \\ \sigma_{01}(\lambda u, \lambda \omega_1, \lambda \omega_2) &= \sigma_{01}(u, \omega_1, \omega_2), \\ \sigma_{10}(\lambda u, \lambda \omega_1, \lambda \omega_2) &= \sigma_{10}(u, \omega_1, \omega_2). \end{aligned} \quad (8)$$